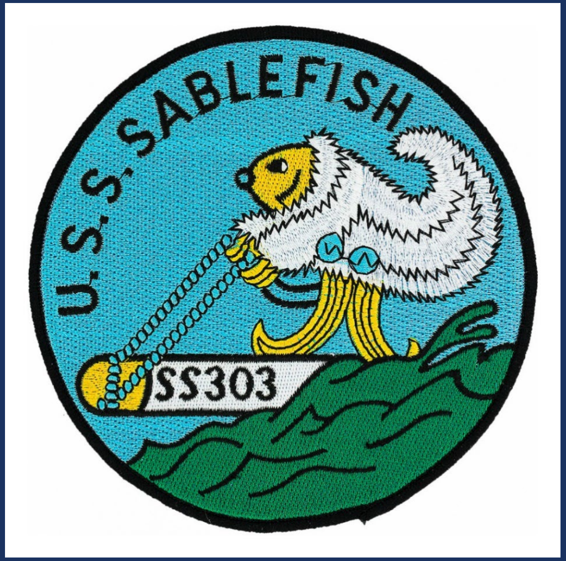
Sablefish Assessment: Available Data

2026 CIE Review

Daniel Goethel and Matt Cheng

Contributors: Kevin Siwicke and Katy Echave

June 2026



Agenda

Tuesday, June 16, 2026

A. **Welcome and Introductions (30min)**
B. **Sablefish Background (Management, Biology, and Data Inputs; 3hrs)**
C. **General Assessment Model Structure (ToR 1; 3hrs)**

Wednesday, June 17, 2026

A. **Reconvene and Review (30min)**
B. **Summary of Closed Loop Simulations (3hrs)**
C. **Comparing Assessment Performance and Diagnostics (ToRs 1/2; 3hrs)**

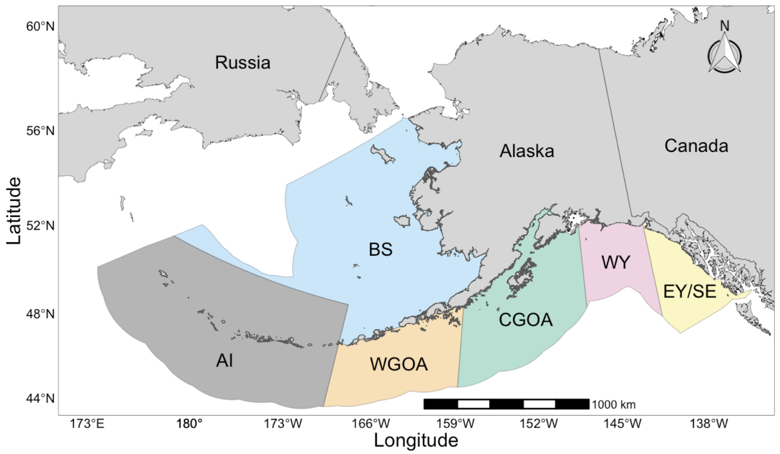
Thursday, June 18, 2026

A. **Reconvene and Review (30min)**
B. **Comparing Assessment Performance and Diagnostics *Cont'd* (ToR 1/2; 3hrs)**
C. **Spatial Model Configurations and Diagnostics (ToR 3; 3hrs)**

ADJOURN

Issues to Consider: Data Inputs

- What is the best approach for constructing a single region NOAA Longline Survey?
  - Is the 'carryover' extrapolation method appropriate?
- Are all data truly representative of the Alaska-wide stock?
  - Does integrating CPUE or GOA trawl survey improve model performance?
- When do length composition data improve the model?
  - If age compositions exist from a given data source, should length compositions be removed?
- Are regional age composition sample sizes sufficient for use in the fleets-as-areas (FAA) model?



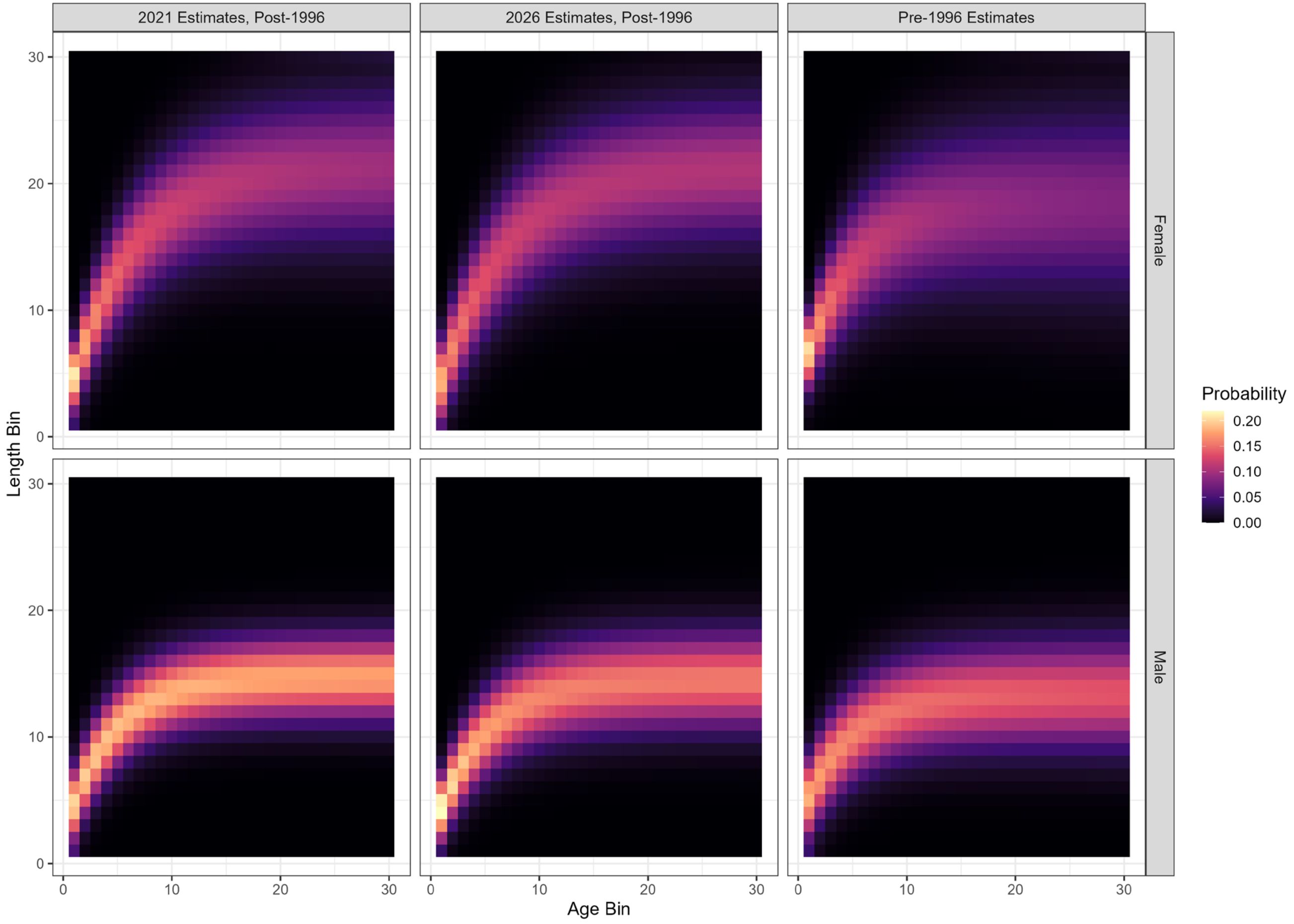
Data Summary

- New data for 2026 in **bold**
- Whale depredation by area held constant at 2022 levels
- Data removed since 2025 (Model *25.12*) is ~~**text**~~

Source	Data	Years
Fixed Gear Fisheries	Catch	1960 – **2025**
Trawl Fisheries	Catch	1960 – **2025**
Non-Commercial Catch	Catch	1977 – **2025**
~~Japanese Longline Fishery~~	~~Catch-per-Unit-Effort (CPUE)~~	~~1964 – 1981~~
U.S. Fixed Gear Fisheries	~~CPUE~~	~~1995 – 2022~~
	~~Length~~	~~1990 – 1998~~
	Age	1999 – **2024**
U.S. Trawl Fisheries	Length	1990,1991,1999, 2005 – **2024**
Japan-U.S. Cooperative Longline Survey	RPNs, Length	1979 - 1994
	Age	1981, 1985, 1987, 1989, 1991, 1993
NOAA Domestic Longline Survey	RPNs	1990 – 2023, **2025**
	~~Length~~	~~1990 – 1995~~
	Age	1996 – 2023

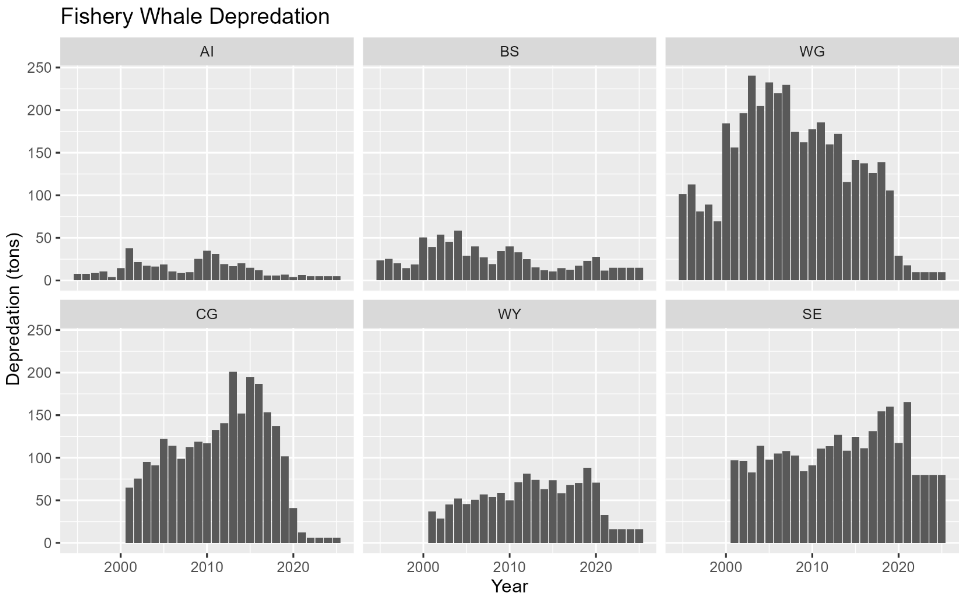
Biological Inputs

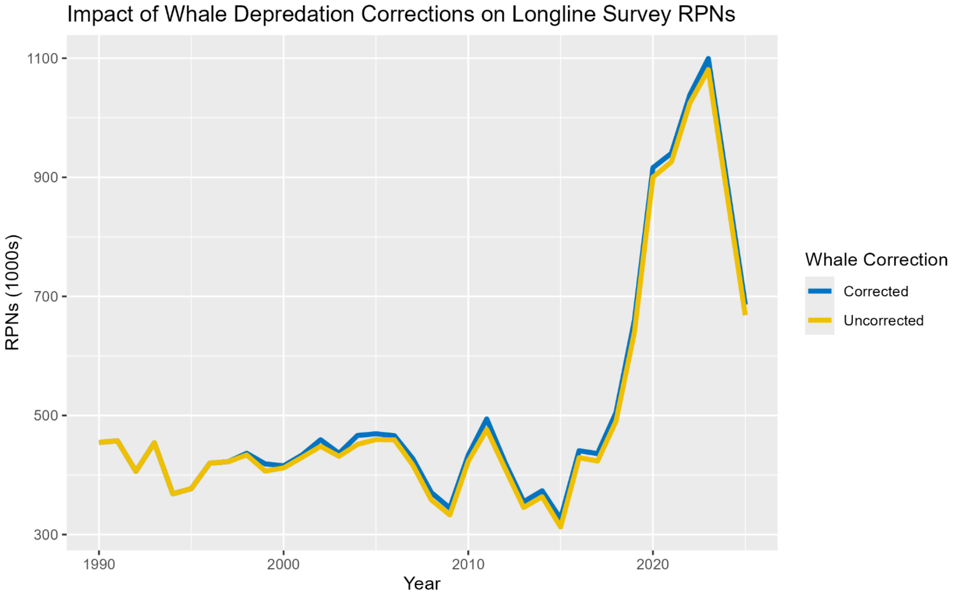
- Updated size-age transition matrix and weight-at-age in 2026 with NOAA LLS age-length-weight pairs through 2023
- Ageing error matrix based on known age otoliths
- Maturity-at-age based on histological samples
- Natural mortality prior ~$N(\ln(0.085), 0.1^2)$ based on longevity and tagging estimates



Whale Depredation

- Fishery whale depredation no longer updated annually (full update in 2022)
  - Limited depredation due to rapid transition to pot gear (no assumed depredation)
  - Totals ~ 131 mt annually
- Sperm whale depredation effect on longline survey is low and the area RPNs are adjusted using a fixed scalar of 1.18 for *affected* stations (relatively small number of stations)
  - Killer whale depredated stations are removed from the RPNs

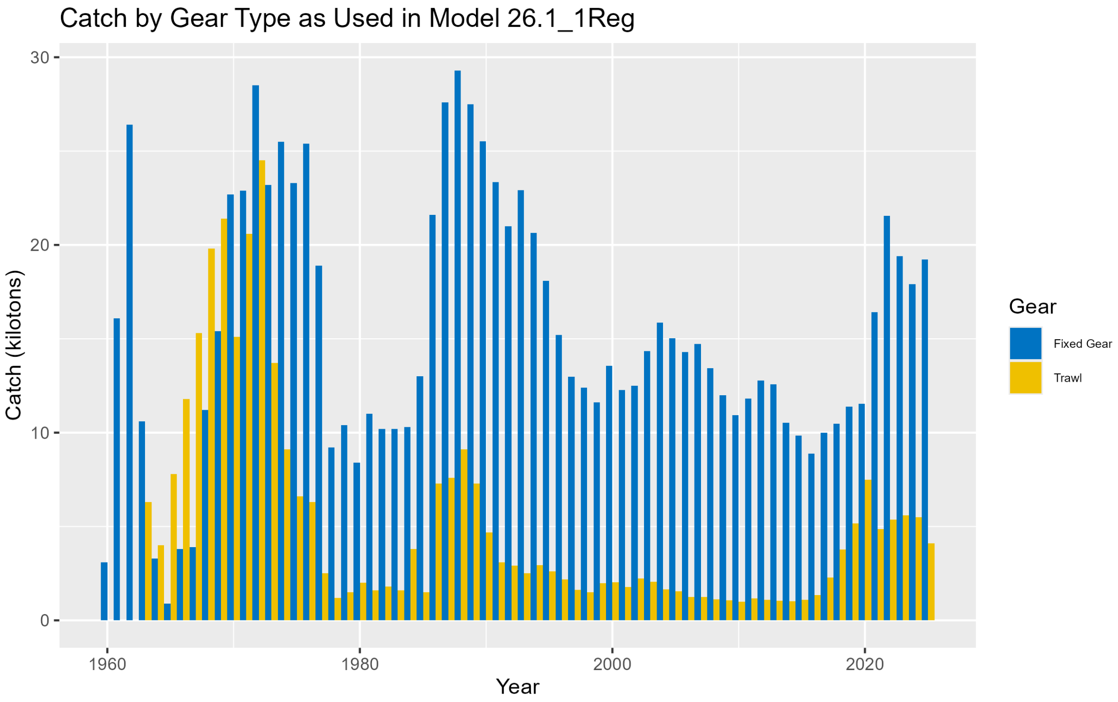




*Note figure does not include non-surveyed area extrapolations

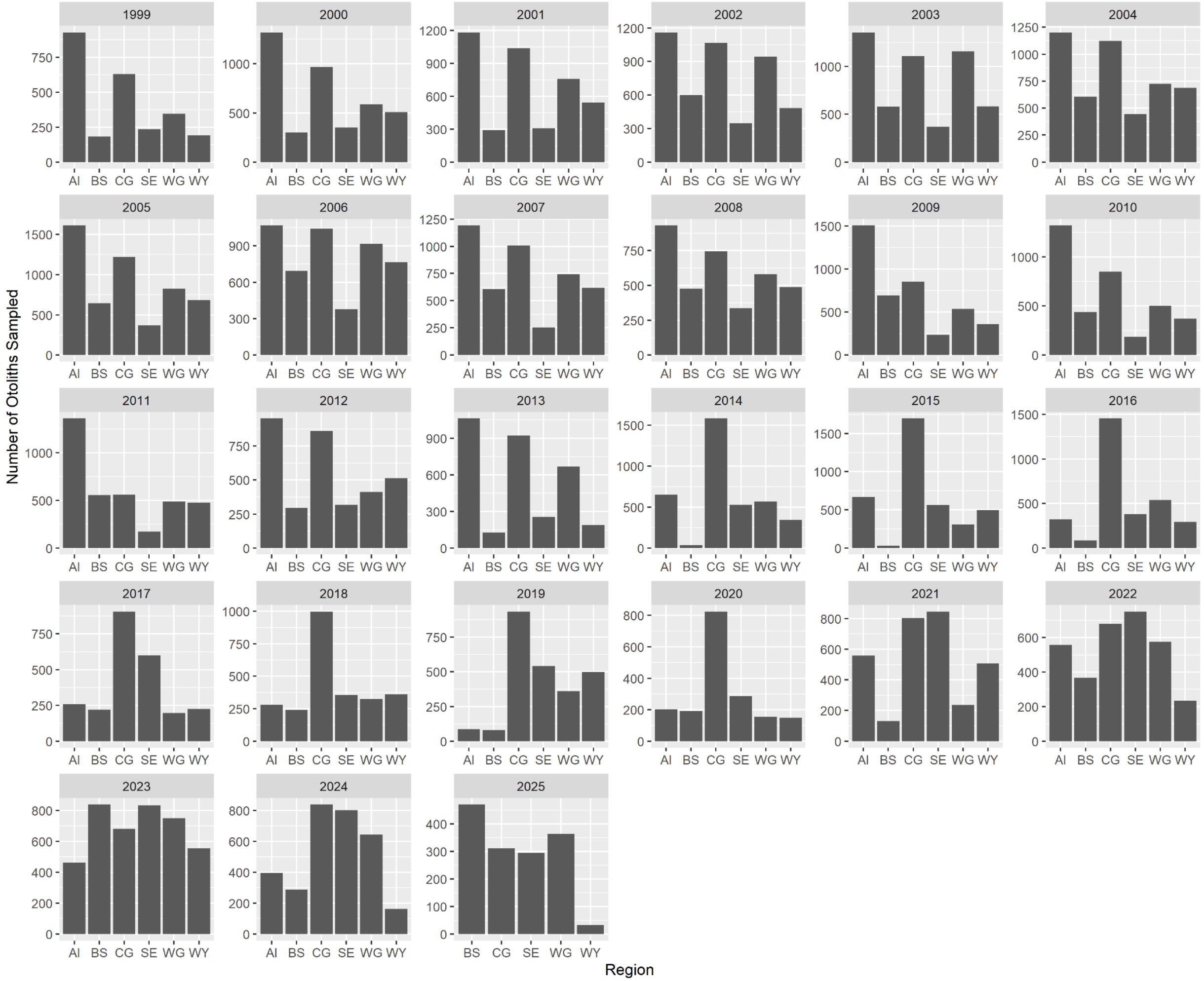
Catch

- Catch is aggregated to two fleets (fixed gear and trawl gear) for the assessment
- Catch is summed across regions and gear types within fleets
- Combines Japanese and domestic fleets
- Fixed gear fleet includes hook-and-line and pot gear
  - Increased from ~15% in 2017 to ~85% pot gear in 2025
  - Additional catch added from:
    - Non-commercial catch (~200 -- 500 mt annually), primarily from LLS
    - Whale depredation (~131 mt annually)
- Trawl fleet is 10 – 30% of domestic total catch and includes:
  - Pelagic gear -- primarily eastern Bering Sea (EBS) shelf pollock fleet
  - Bottom trawl gear – in AI+GOA where occasionally have set specific sablefish targeting (trip 'top-off')



Fixed Gear Fishery Age Compositions

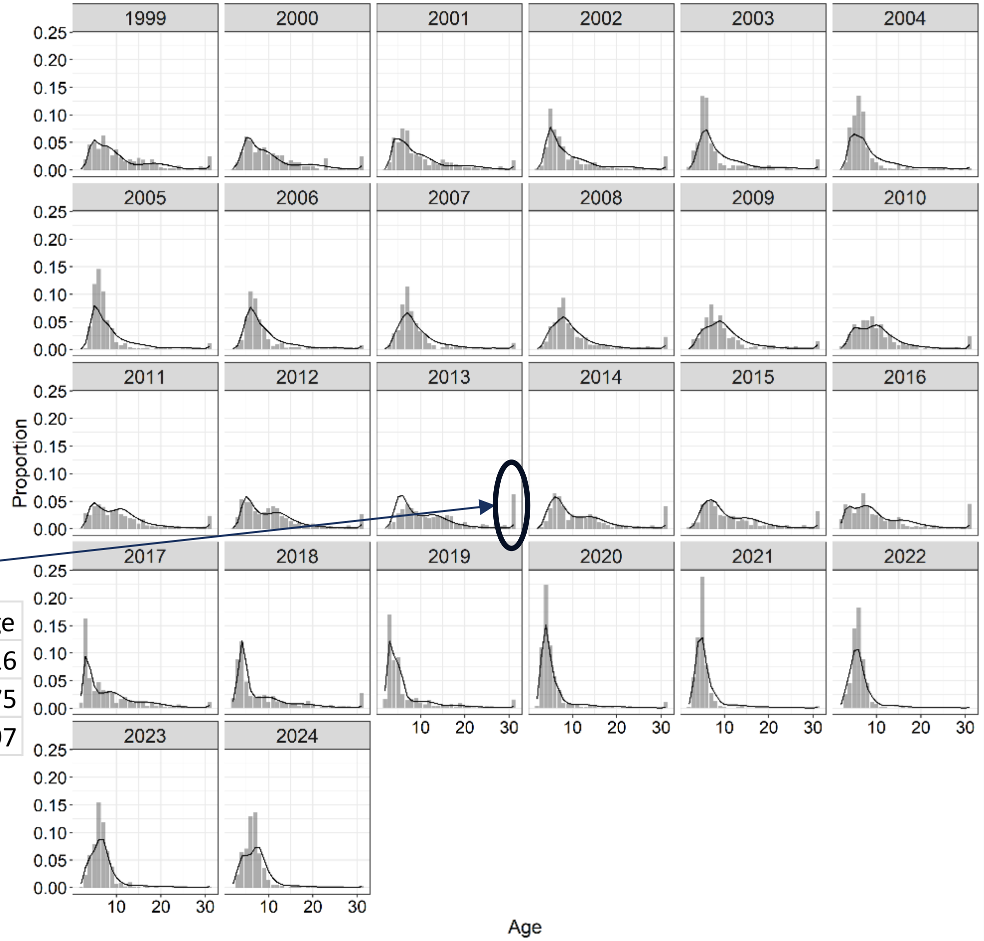
- Random otolith collection by observers (~2,000 – 5,000 total specimens) since 1999
- ~1,200 otoliths aged per year by AFSC
  - Available on a one-year lag to the assessment
- Post collection a random sampling protocol (e.g., across size, sex, and haul) is implemented, which selects otoliths for aging from all otoliths collected
  - Aim for ~200 per each of six management regions
  - Strong reductions in recent years due to Electronic Monitoring and changes in observer coverage



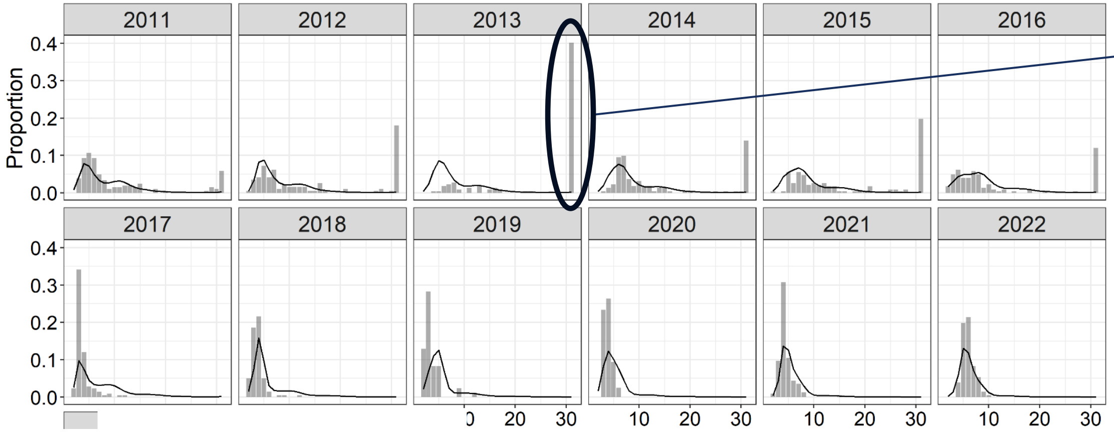
Fixed Gear Fishery Age Compositions

- Annual Alaska-wide catch-weighted (in numbers) age compositions produced (mean across regions)
  - Use catch in numbers to better represent recruitment
- Produce representative compositions from across Alaska without overweighting regions with low catch
  - **Emphasizes signals from GOA (high catch, high sample sizes)**
- …but the AI plus group is an enigma….

Alaska-wide Female Age Comps



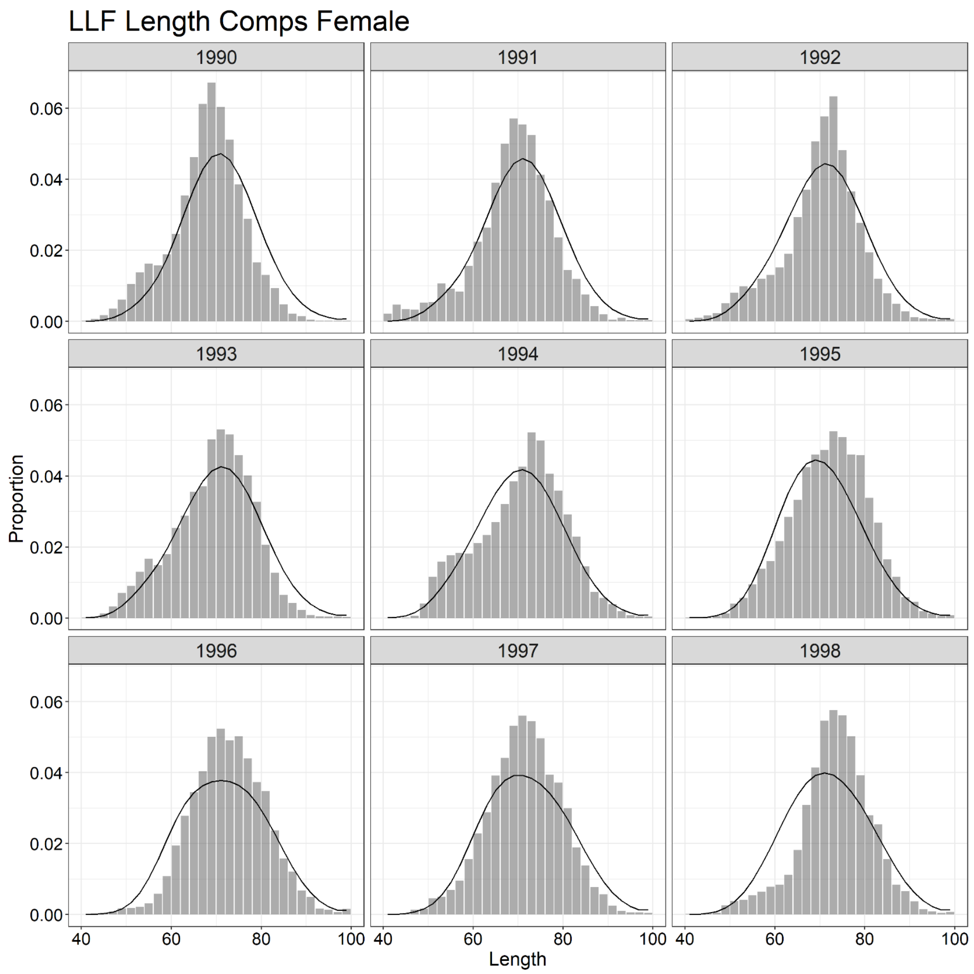
AI Age Comps



Region	Mean age
East AI	7.705716
Cent AI	12.75075
West AI	40.58297

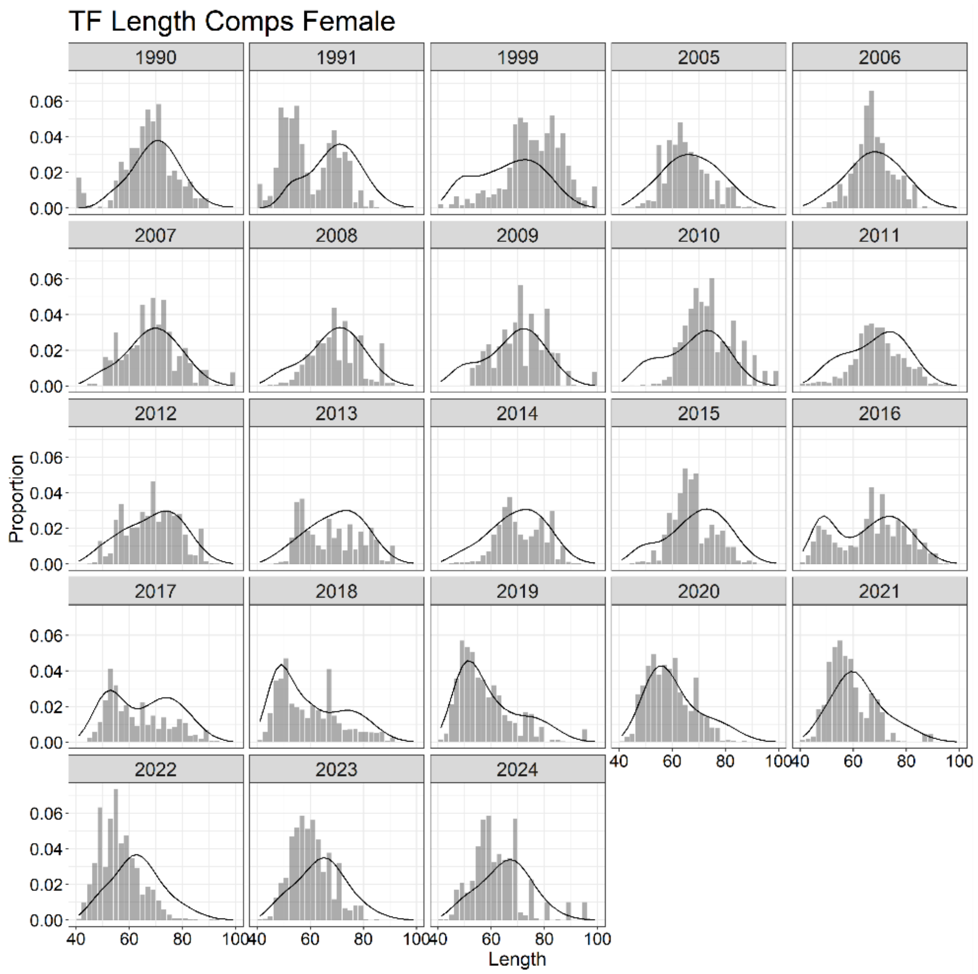
Fixed Gear Fishery Length Compositions (Not Used)

- Observer collected length samples available since 1990
- ~12,000 – 30,000 per year
- Available on one year lag for assessment
- Annual Alaska-wide catch-weighted length compositions produced
- Not currently used since already fitting age compositions
  - Increased potential for process error in fitting lengths in an age-structured assessment



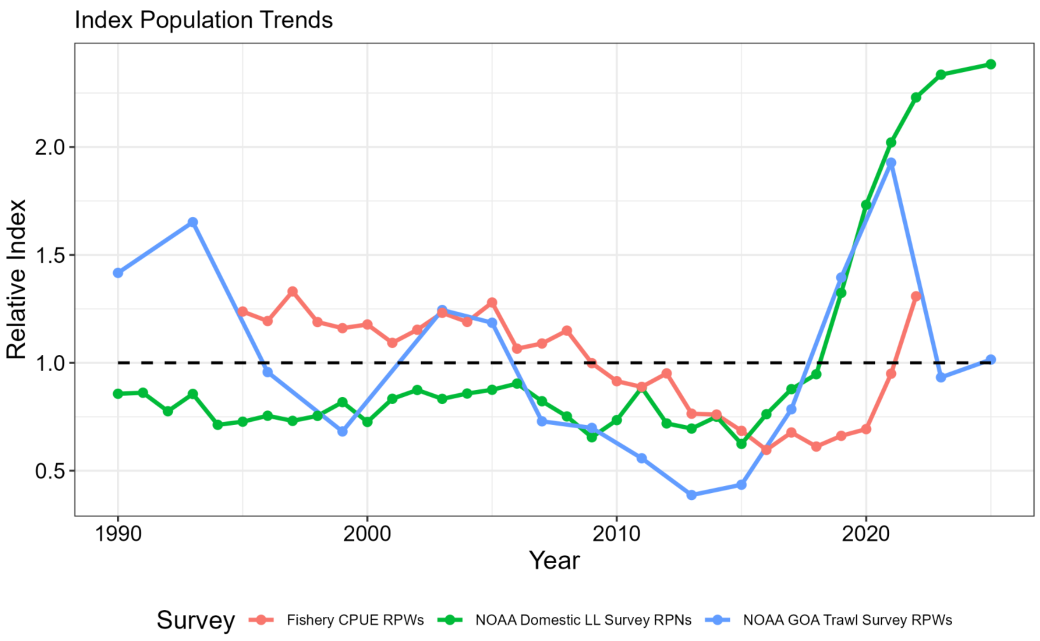
Trawl Fishery Length Compositions

- Observer collected length samples available since 1990
- ~500 – 6,000 per year
- Available on one year lag for assessment
- Annual Alaska-wide catch-weighted length compositions produced
- Often bi-modal due to the aggregation across gears that have distinct interactions with sablefish
    - Pelagic trawl primarily interacts with juvenile sablefish
- No age data available from trawl fleets



Indices of Abundance/Biomass

- Three indices available to inform sablefish population trends:
  - NOAA Longline Survey – sablefish directed survey sampling across Alaska
  - NOAA Gulf of Alaska Trawl Survey – multispecies survey sampling juvenile sablefish in the GOA
  - Fixed Gear Fishery CPUE – standardized CPUE index from observer and logbook data, primarily in GOA



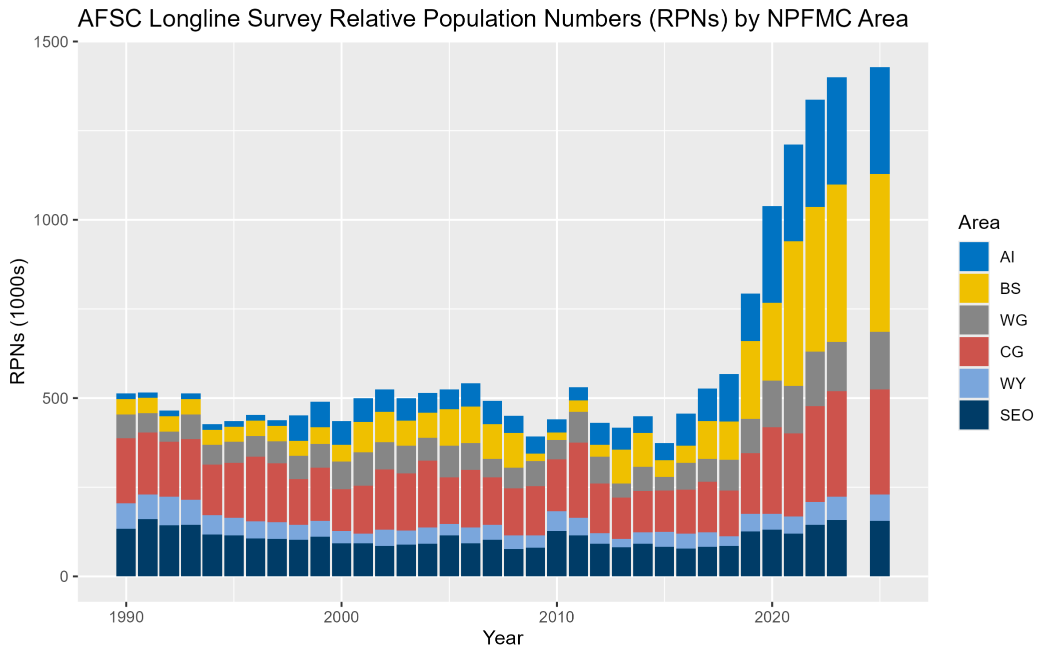
LL Survey: Slight Increase

CPUE Index: No Update

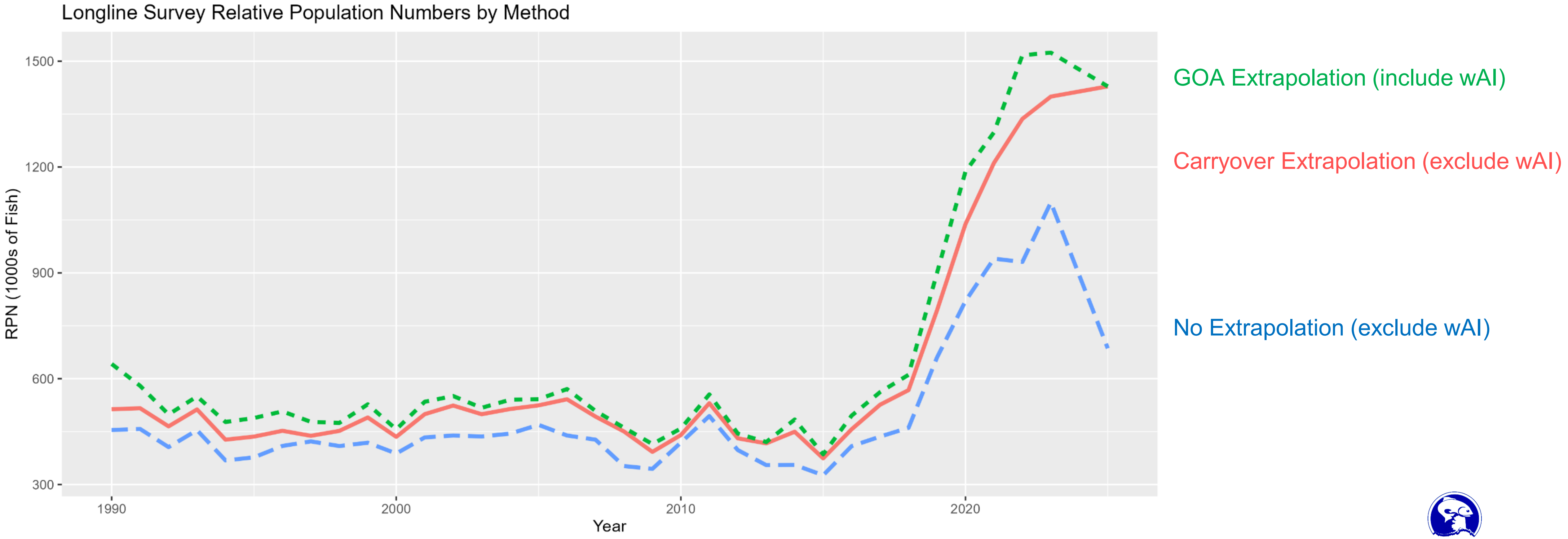
Trawl Survey: Slight Increase

NOAA Longline Survey Index

- Extrapolated relative population numbers (RPNs) are required to ensure consistent AK-wide scaling
  - Prior to 2025 change in GOA used to extrapolate BS or AI from previous year (**No Longer Feasible**)
  - Since 2025 'carryover' approach maintains the previous area-specific RPN for any unsampled area
    - Extrapolating ~50% of the RPNs under new survey design, given recent sablefish distribution and catch rates in BS+AI
- Prior to 2026 utilized extrapolations for western AI (wAI) stations based on Japanese LLS data and outdated area sizes
  - Removed in 2026 resulting in ~2 – 10% index reduction

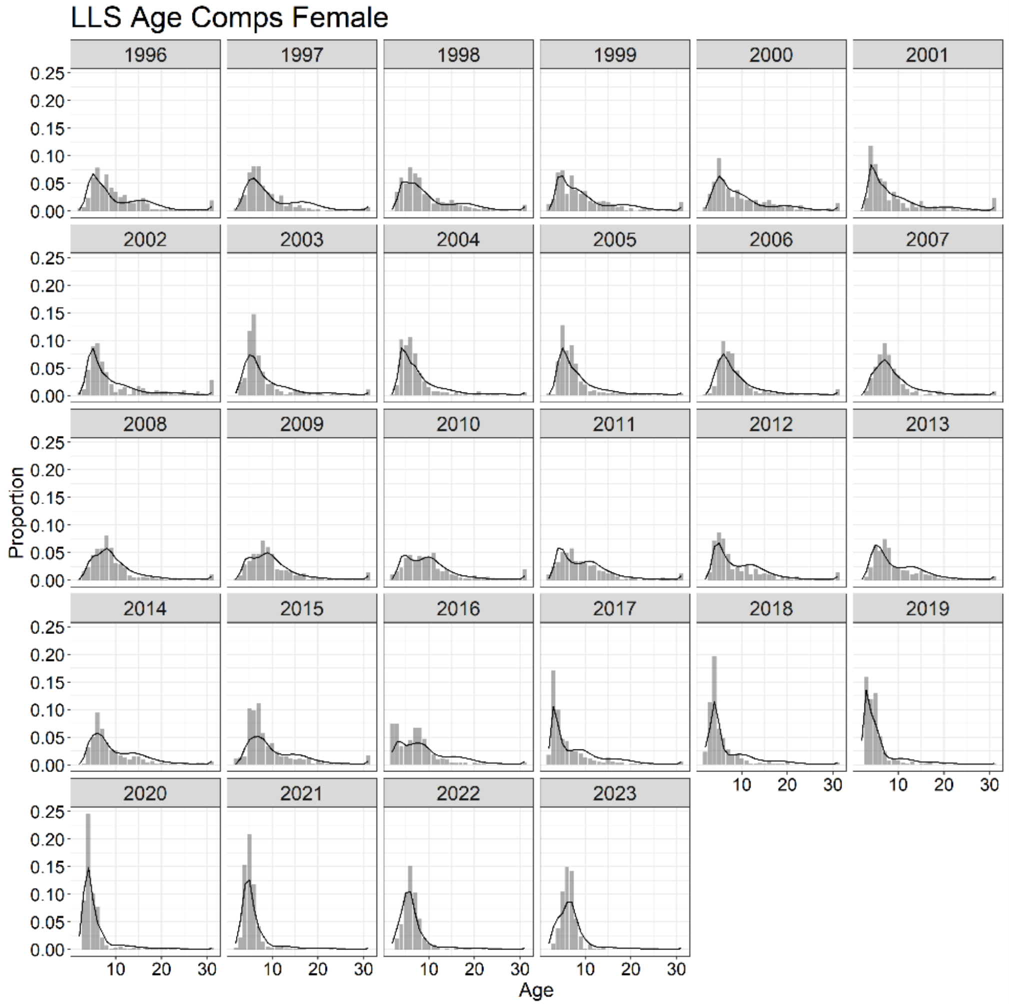


NOAA Longline Survey Index



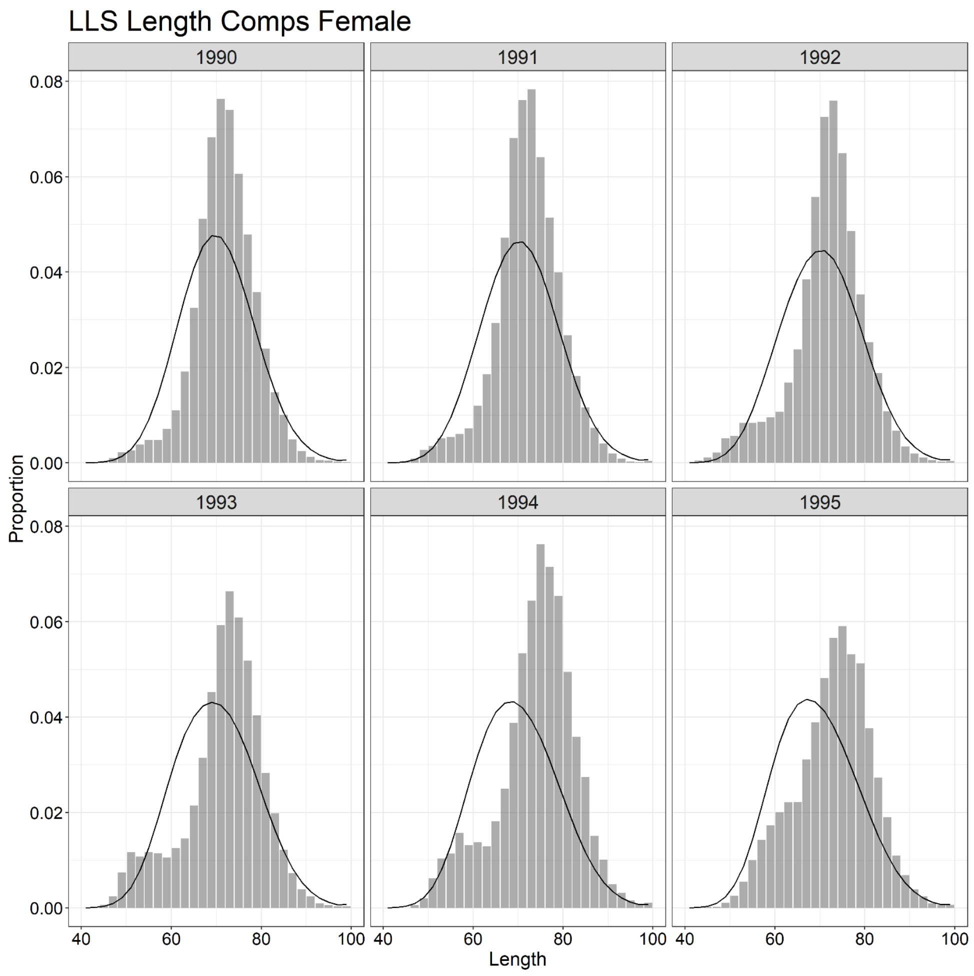
NOAA Longline Survey Age Compositions

- Random otolith collection by scientists (~1,500 – 4,000 total specimens) since 1996
- ~1,200 otoliths aged per year, but only ~900 from stations used in the index
  - Available on a one-year lag to the assessment
- Random sampling:
  - Before 2025 aimed for ~200 each from EG/CG/WG and 300 from BS or AI
  - Since 2025 aim for ~300 each from EG/CG/WG in odd years and 600 each from BS/AI in even years
- Annual Alaska-wide RPN-weighted age compositions produced



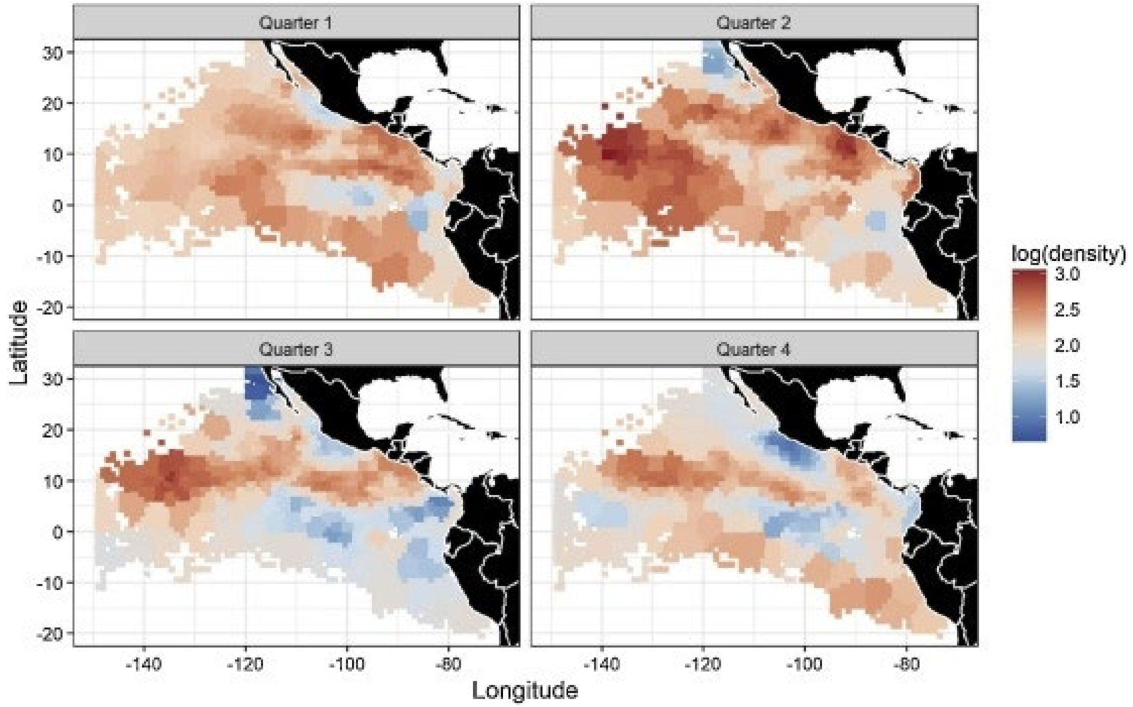
NOAA Longline Survey Length Compositions (Not Used)

- Observer collected length samples available since 1990
- ~40,000 – 90,000 per year
- Available in-year for assessment
- Annual Alaska-wide RPN-weighted length compositions produced
- Not currently used in assessment since already fitting age compositions



NOAA Longline Survey Index – Alternatives?

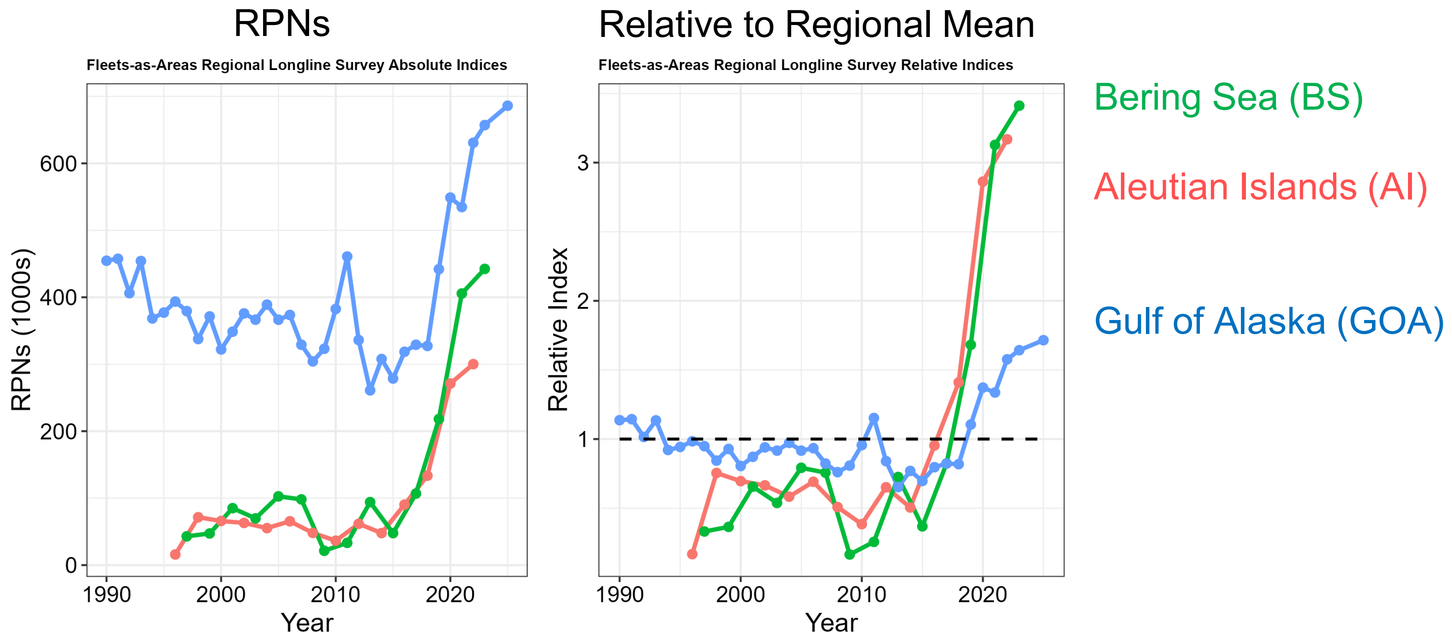
- Model-based methods
  - Unlikely to be effective if certain areas have never been sampled simultaneously or if extrapolating over large areas
  - Added difficulty interpolating given nature of data (fixed stations/longline gear)
- Ad-hoc imputation
  - Fill in missing values with historical values
  - Risks biases if historical values are not reflective of current conditions
- Regional indices and compositions
  - Fit data at the ~scale at which they were collected
  - Data sparsity issues for lesser sampled regions



Xu et al 2019

NOAA Longline Survey Regional Indices

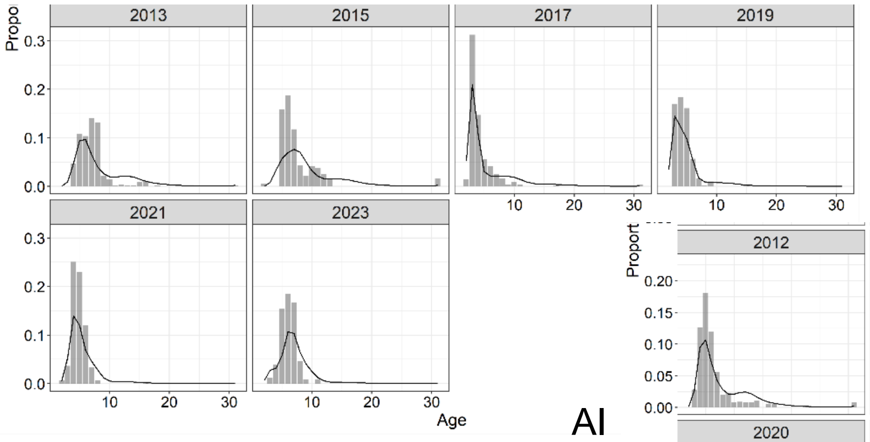
- For fleets-as-areas (FAA) model, disaggregated indices (BS, AI, GOA) were produced
  - GOA aggregated across associated regions (EG/CG/WG)
- No extrapolation necessary (unsampled regions treated as missing data; no placeholder)
- Dropped western AI extrapolated stations (matching single region index)



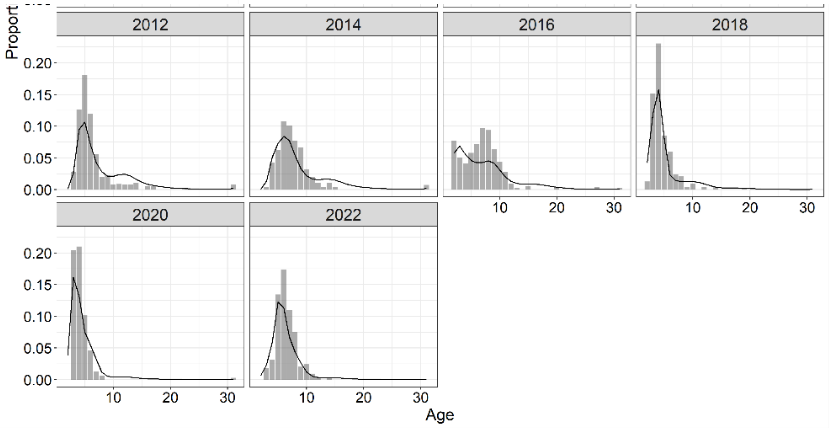
NOAA Longline Survey Regional Age Compositions

- For BS and AI, raw age compositions used (no weighting)
- For GOA, RPN-weighted age compositions produced
- Sample sizes in BS/AI are half of those in GOA (300 vs. 600)
  - AK-wide and GOA age compositions generally align

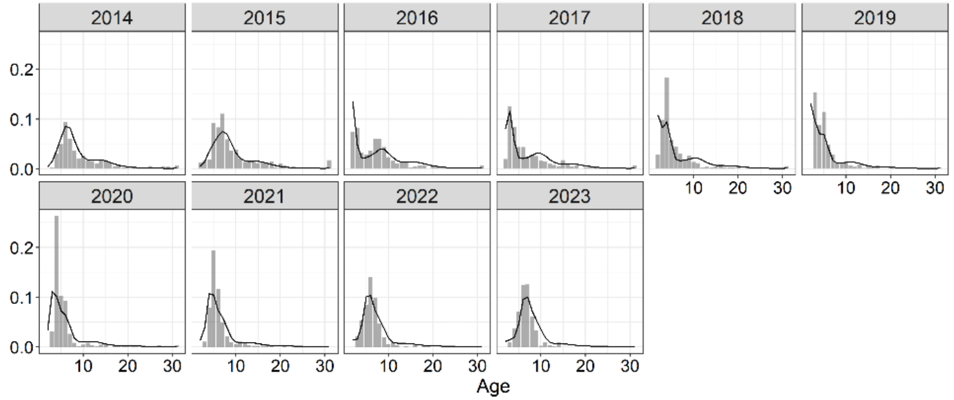
BS



AI

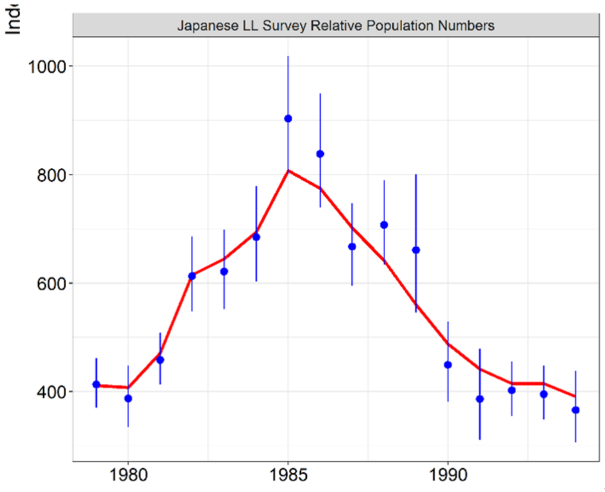


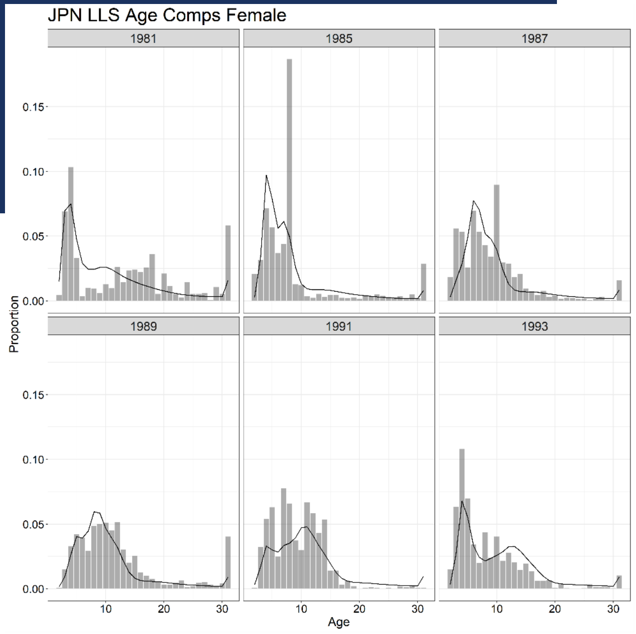
GOA

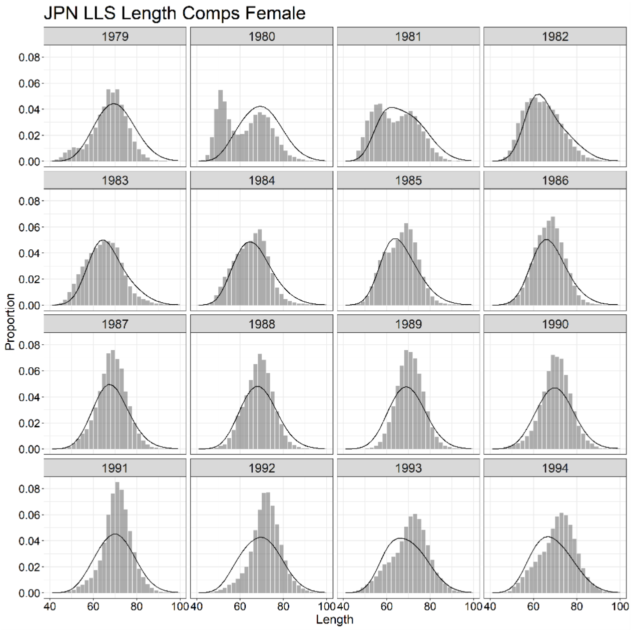


Japanese Longline Survey

- NOAA Longline Survey essentially adopted similar methods as historical Japanese LLS
- Japanese LLS RPNs available from 1979 – 1994
- Length-stratified age compositions available sporadically
  - ~800 – 1,700 samples
- Length compositions available annually
  - ~18,000 – 120,000 samples
  - Integrated into assessment to enable estimation of selectivity
- AK-wide RPN-weighted age/length compositions produced

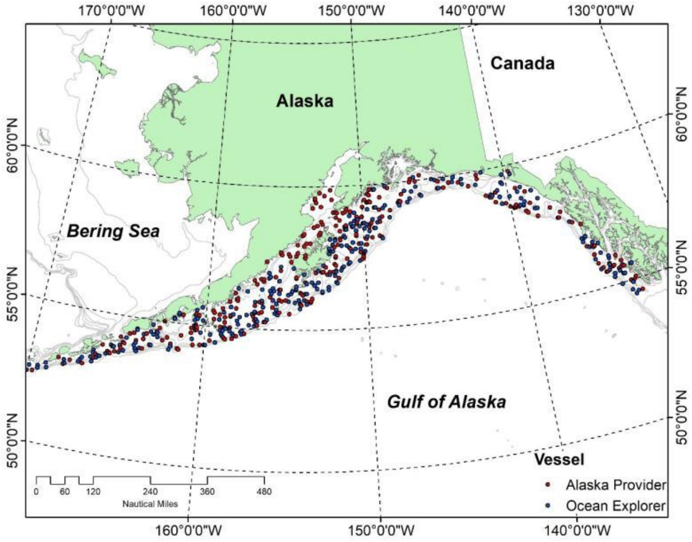


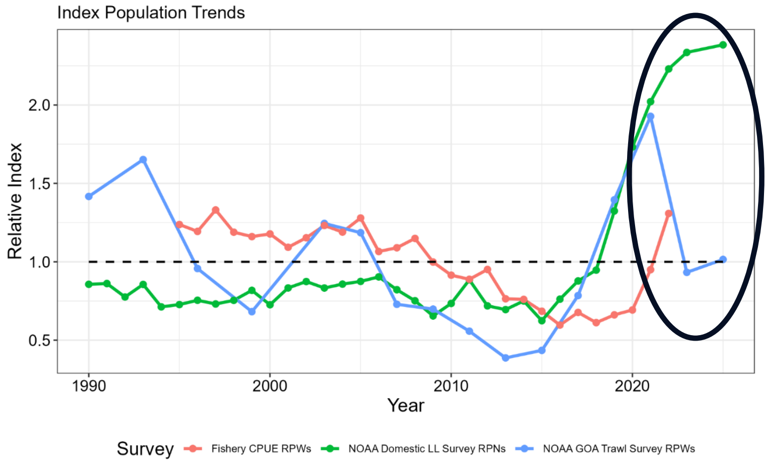




Trawl Survey Index and Length Compositions (Not Used)

- Multispecies survey of the continental shelf to 500m
  - Random stratified survey design
  - Triennial (1990 – 2003), biennial (2003 – Present)
  - 1,000 – 8,000 length samples taken per survey (no otoliths aged)
- Data Issues (Representativeness):
  - Only covers the GOA
    - BS and AI trawl surveys have shorter time series and more sporadic catch of sablefish
  - Only consistently surveys < 500m, which is shallower than prime sablefish habitat
  - Primarily catches only juvenile (smaller) sablefish



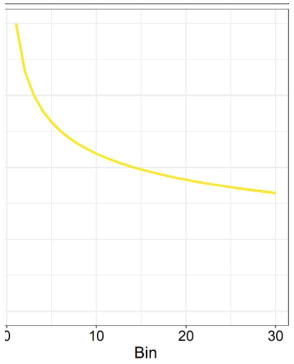


Trawl survey abundance indices were not previously used in the sablefish assessment because they were not considered good indicators of the sablefish relative abundance. However, there is a long time series of data available and given the trawl survey's ability to sample smaller fish, it may be a better indicator of recruitment than the longline survey. Hanselman et al., 2007

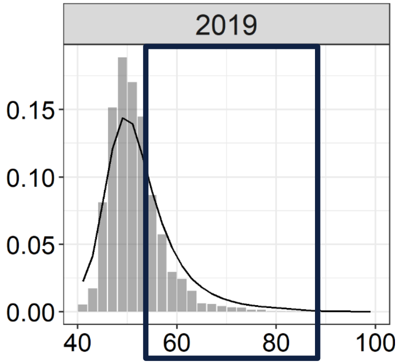
Trawl Survey Index and Length Compositions (Not Used)

- Integrated in 2007 as a recruitment index
  - **Removed in 2025**
- Appropriate selectivity is difficult to define
- Has high influence on recent recruitment estimates in the assessment
  - Increases variability in recruitment signals since it is assumed to select ~100% of age-2 fish
- Often contradicts LLS composition data leading to recruitment retrospective patterns
  - Recent signals differ greatly from LLS (e.g., 2023 and 2025 declines)

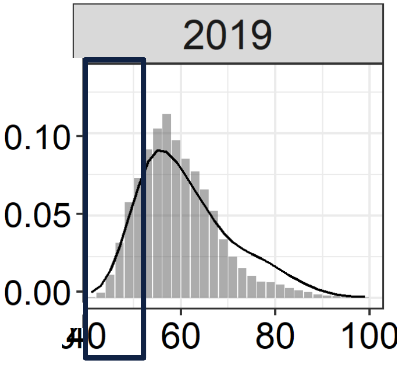
Male Trawl Survey Selectivity



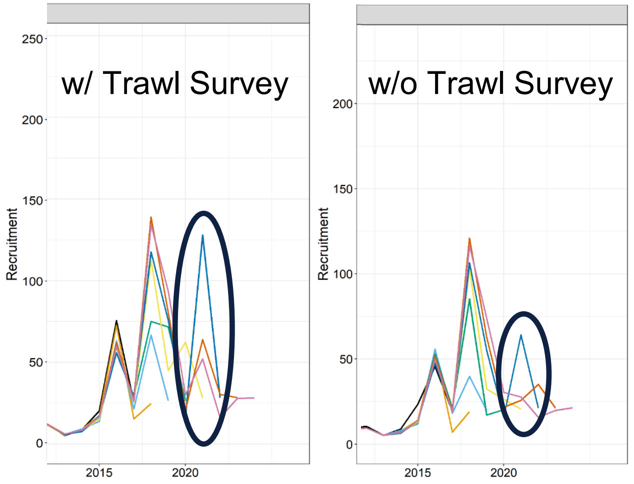
Trawl Survey Female Lengths



Longline Survey Female Lengths



Recruitment Retrospective Patterns

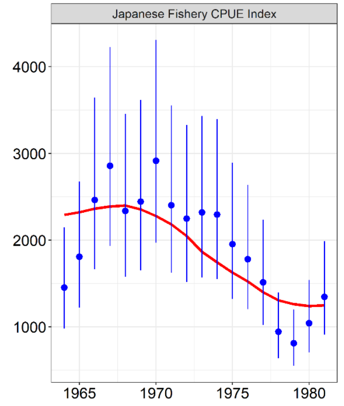


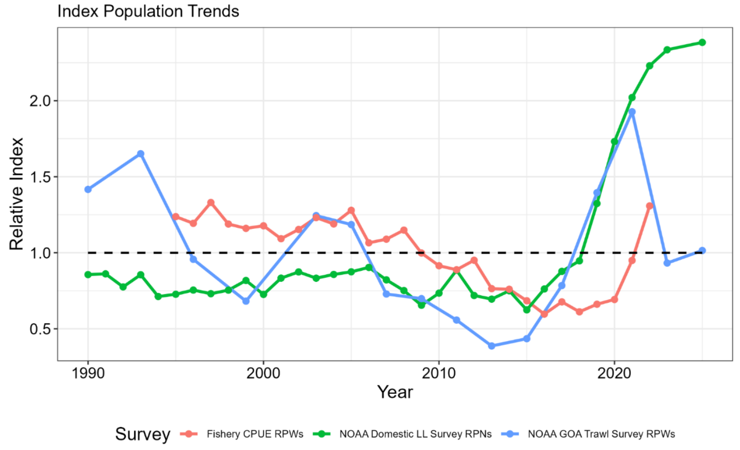
Fixed Gear CPUE (Not Used)

- Nominal Japanese longline CPUE (1964 – 1981)
  - Reported in Sasaki (1985)
- Standardized domestic fixed gear CPUE (1995 – 2022)
  - Records of catch and effort for sablefish targeting vessels are collected by observers and reported in logbooks, primarily in GOA
  - A combined fixed gear standardized CPUE index was developed
  - Can no longer be updated (logbooks not provided to NOAA)
  - Trends mostly align with LLS RPNs, with lag (represents mature biomass)
- Data and assessment issues:
  - Difficulty standardizing across management regimes
  - Assume a high CV compared to LLS, so provides minimal information to assessment

Standardizing fishery-dependent catch-rate information across gears and data collection programs for Alaska sablefish (*Anoplopoma fimbria*)

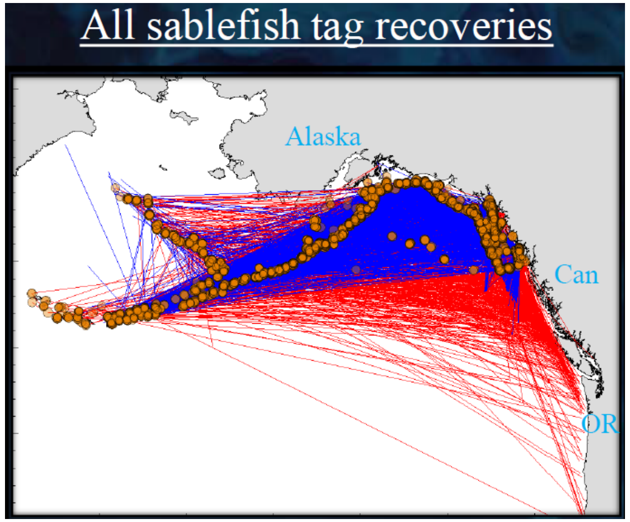
M. L. H. Cheng[1,*], C. J. Rodgveller[2], J. A. Langan[1], and C. J. Cunningham[1]





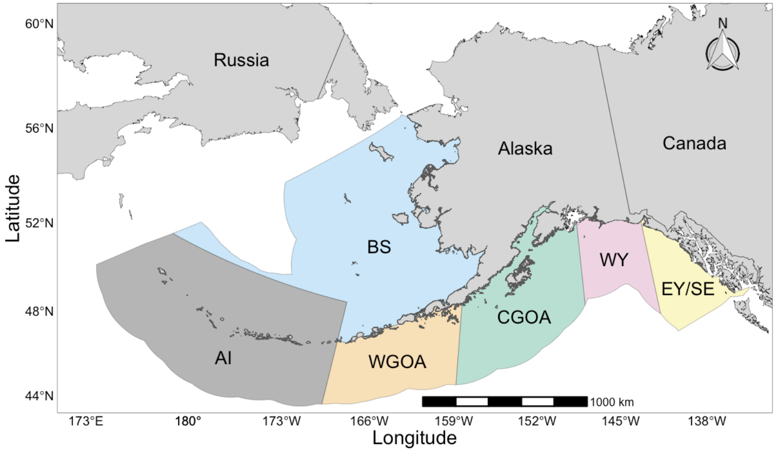
Mark Recapture Data

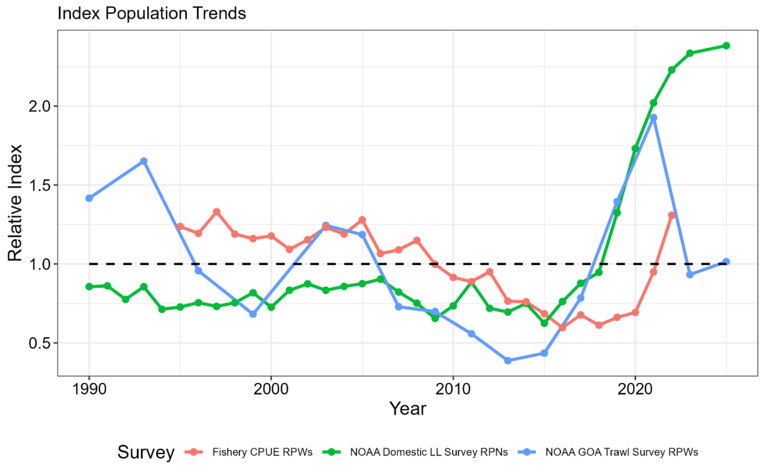
- Tagging with conventional tags since 1970s (~413,000 releases)
  - ~ 3,500 adult sablefish tagged per year on the LLS
  - ~ 100 – 1,000 juvenile (age-1) sablefish tagged per year in nursery areas in the EG
  - Data collected includes location, date, length, depth, and gear
- Recaptures (~38,500):
  - Data collected includes location, time at liberty, depth, size, and otolith (survey recaptures)
  - Average distance between release and recapture is 340 nm
  - Furthest recovery was 2,586 nm
  - Fastest traveled 116 nm in 6 days
  - Longest time at liberty >39 yrs
- Uses:
  - Estimating growth, ageing error (known-age studies), movement rates, catch apportionment, and movement patterns
  - Integrated directly into the spatial assessment models

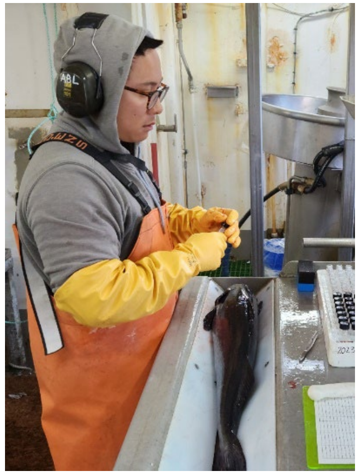


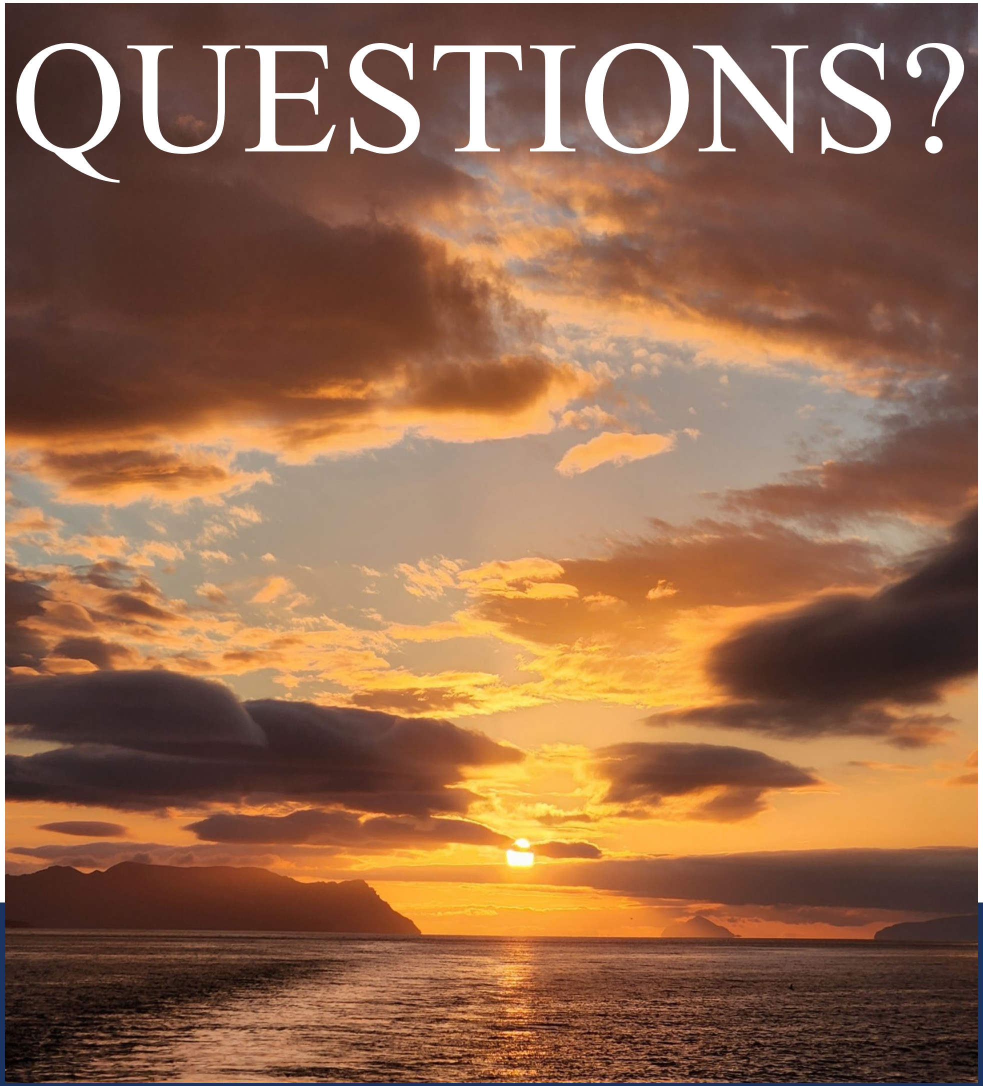
Key Considerations for Assessment

- LLS represents best scientific information available on sablefish trends, but difficult to account for survey changes
  - Is it better to fit extrapolated data or disaggregated data (with lower composition sample sizes)?
- Should alternative indices be integrated (CPUE or trawl survey), if they reflect dynamics of a subset of the population (e.g., GOA-only, juveniles or mature fish)
- Need to integrate size-age transition matrix to fit trawl fishery and Japanese LLS length compositions
  - Is there benefit (or increased process error) from fitting age and length compositions simultaneously (e.g., for LLS and fixed gear fishery)?








QUESTIONS?